[scan 25]

B)

If 2 invertible, we will have therefore

$$\mathrm{Sym}^*(V_J|W_J)^{\mathfrak{S}_{J|3}} \simeq \mathcal{O}[$$

Remark

Since γ_2 vanishes at the P_ω, γ_3 at the Q_i, it follows that there is no invariant section of an $\mathcal{O}(n)^*$ for over Σ_J^* in degree $<$ for $n \leq 6$, hence that the order of $\mathrm{cl}(\mathcal{O}(1))$ in $H^1(\Sigma_J, \mathfrak{S}_J; \mathbb{G}_m)$ is indeed 6, which proves that $\mathcal{O}(1)$ is a generator of the mixed H^1. 24